

AI-Powered Routing Reduced Audit Intake Friction Without Weakening Controls

Operational intelligence case built from anonymized real work in a regulated financial-services context.

657+ items

97%+ accuracy

24+ hrs saved

Anonymized real work

Work Type

Anonymized real work

657+

audit requirements managed

97%+

assignment accuracy

24+ hrs

manual work removed weekly

<3%

misroute escalation rate

The legacy process was expensive because judgment and mechanics were bundled together

The bottleneck was routing and prioritization, not the substantive audit work itself.

Key points

- Every incoming requirement required manual reading, categorization, and forwarding.
- Experienced staff handled mechanical sorting work that did not justify their time.
- Misroutes and delayed ownership created audit friction and avoidable rework.

Before-state economics

24+ hours of senior staff time were spent weekly on a task that should have been structured, not artisanal.

The workflow was rebuilt as a three-layer control system

The design accelerated routine cases while reserving ambiguity for human review.

Classification

Interprets the requirement, extracts key entities, and assigns a primary category.

Routing logic

Maps the case to the accountable owner through an editable rules layer.

Validation

Uses confidence thresholds to separate auto-routing from human escalation.

The architecture was designed to remain editable under changing audit needs

Separating interpretation, routing logic, review thresholds, and logging is what made the workflow maintainable.

Layer	Primary role	Editable element	Risk reduced
Interpretation	Read and classify	Examples and categories	Inconsistent intake reading
Routing matrix	Assign owner	Rules table	Hard-coded routing rigidity
Confidence review	Escalate ambiguity	Threshold logic	False certainty
Logging	Preserve traceability	Case history	Opaque automation behavior

The value is not just speed. It is a workflow that can evolve without losing control logic.

The results changed both throughput and control quality

The strongest result is that speed improved alongside consistency, not instead of it.

Metric	Before	After	Change
Processing time	15-20 min	30-45 sec	95% reduction
Weekly manual effort	24+ hrs	<2 hrs	92% reduction
Assignment accuracy	~88%	97%+	+9pp
Misroute escalation	~12%	<3%	Material reduction
Response cycle	24-48 hrs	<4 hrs	Much faster

Metrics reflect workflow handling quality rather than model-lab benchmark scoring.

The governance layer is what made the workflow usable in a regulated setting

The design did not pretend uncertainty did not exist. It operationalized it.

Key points

- High-confidence items were auto-routed to preserve throughput.
- Low-confidence items were escalated to human review instead of being forced through the system.
- Routing logic was editable without code changes, which preserved maintainability and auditability.

Why this matters

The asset proves Santiago can design AI-enabled workflows with control logic, not just prompts.

The same design pattern transfers to other high-volume judgment workflows

What generalizes is the operating architecture: classify, route, review ambiguity, and preserve traceability.

Document triage

Useful where many incoming requests need fast assignment without losing context.

Control-sensitive operations

Relevant when exceptions and reviewability matter as much as speed.

Human-in-the-loop AI

Strong fit for systems where automation should narrow labor, not remove judgment.

This is proof of execution under constraints

The case is strongest not because it uses AI, but because it shows AI connected to workflow design, governance, and measurable business impact.

Takeaways

- Relevant for workflow automation, document triage, compliance operations, and operational intelligence work.
- For recruiter review or project discussion, email santiago.garces.blevine@gmail.com. Include the role or problem, timeline, and expected deliverable format.

Contact

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Market Research | Competitive Intelligence | AI
Workflow Execution

Based on real operating work in a regulated financial-services environment. Sensitive context is anonymized; metrics are preserved.

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